

Yolov11 Garbage Recognition Broadcast

1. Function Description

Voice broadcast of the recognized garbage name and garbage type.

2. Startup and Operation

2.1 Startup

First, you need to start the ROS node service. Open the terminal and enter the following:

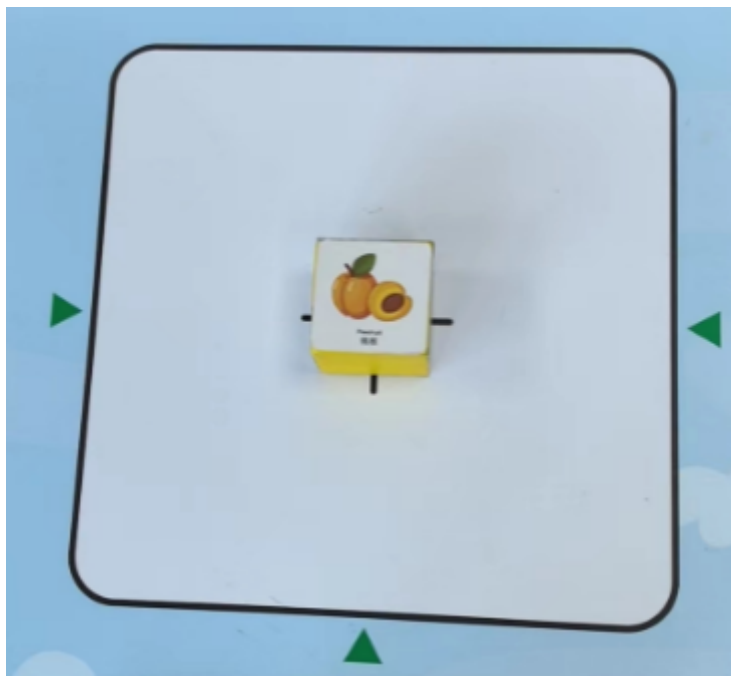
```
ros2 run dofbot_info kinemarics_dofbot
```

Then open another terminal and enter the following:

```
python3 ~/dofbot_voice/scripts/garbage_broadcast.py
```

2.2 Operation Steps

Place the garbage label block in the middle of the image. Say "Hi, yahboom" to the voice module. The voice module replies "I'm here" indicating successful wake-up. Then say "What garbage is this?". After the program recognizes it, the voice module will broadcast what garbage it is and the garbage type. Taking the figure below as an example, according to the recognition result, it will reply "This is a peach pit, which belongs to dry garbage".



3. Core Code Analysis

Source code path: `~/dofbot_voice/scripts/garbage_broadcast.py`

```
# Import voice broadcast garbage type library
from speech_garbage import speech_garbage
```

```

# Mainly look at the content of this library
# Import garbage recognition library
from garbage_identify import garbage_identify

# Call garbage recognition function, pass the current image as parameter, then
return the processed image and recognition result
self.frame, msg = self.garbage_identify.garbage_run(self.frame)

# Iterate through the recognition result msg, according to the value of name to
determine the garbage name representing number self.garbage_num and type
self.garbage_class
for key, pos in msg.items(): name = key
    if name == "Zip_top_can":                (self.garbage_num,
self.garbage_class) = ('00', '01')
    if name == "old_school_bag":            (self.garbage_num,
self.garbage_class) = ('01', '01')
    if name == "Newspaper":                (self.garbage_num,
self.garbage_class) = ('02', '01')
    if name == "Book":                    (self.garbage_num,
self.garbage_class) = ('03', '01')
    if name == "Toilet_paper":            (self.garbage_num,
self.garbage_class) = ('04', '02')
    if name == "Peach_pit":                (self.garbage_num,
self.garbage_class) = ('05', '02')
    if name == "Cigarette_butts":          (self.garbage_num,
self.garbage_class) = ('06', '02')
    if name == "Disposable_chopsticks":    (self.garbage_num,
self.garbage_class) = ('07', '02')
    if name == "Egg_shell":                (self.garbage_num,
self.garbage_class) = ('08', '03')
    if name == "Apple_core":                (self.garbage_num,
self.garbage_class) = ('09', '03')
    if name == "Watermelon_rind":          (self.garbage_num,
self.garbage_class) = ('10', '03')
    if name == "Fish_bone":                (self.garbage_num,
self.garbage_class) = ('11', '03')
    if name == "Expired_tablets":           (self.garbage_num,
self.garbage_class) = ('12', '04')
    if name == "Expired_cosmetics":        (self.garbage_num,
self.garbage_class) = ('13', '04')
    if name == "Used_batteries":           (self.garbage_num,
self.garbage_class) = ('14', '04')
    if name == "Syringe":                  (self.garbage_num,
self.garbage_class) = ('15', '04')
    if name == "None":                    (self.garbage_num,
self.garbage_class) = ('None', 'None')

# Get voice recognition result
result = mySpeech.speech_read()
# If the current voice recognition result is 94, it means asking what garbage it
is
if result == 94:
    # According to self.garbage_num to broadcast the current garbage name and
garbage type
    if self.garbage_num == '00':
        mySpeech.void_write(94)

    elif self.garbage_num == '01':

```

```
mySpeech.void_write(95)

elif self.garbage_num == '02':
    mySpeech.void_write(96)

elif self.garbage_num == '03':
    mySpeech.void_write(97)

elif self.garbage_num == '04':
    mySpeech.void_write(109)

elif self.garbage_num == '05':
    mySpeech.void_write(108)

elif self.garbage_num == '06':
    mySpeech.void_write(107)

elif self.garbage_num == '07':
    mySpeech.void_write(106)

elif self.garbage_num == '08':
    mySpeech.void_write(105)

elif self.garbage_num == '09':
    mySpeech.void_write(104)

elif self.garbage_num == '10':
    mySpeech.void_write(103)

elif self.garbage_num == '11':
    mySpeech.void_write(102)

elif self.garbage_num == '12':
    mySpeech.void_write(101)

elif self.garbage_num == '13':
    mySpeech.void_write(100)

elif self.garbage_num == '14':
    mySpeech.void_write(99)

elif self.garbage_num == '15':
    mySpeech.void_write(98)
```